

TAKUYA KORIYAMA

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Education.

- 2024-present: PhD student in Econometrics and Statistics, the University of Chicago, Booth School of Business. Advisor: [Tengyuan Liang](#).
- 2022-2024 PhD student in Statistics, Rutgers University. Advisor: [Pierre C. Bellec](#).
- 2018-2022: BE in Mathematical Engineering, the University of Tokyo.

I am broadly interested in the mathematical theory and algorithms underlying statistics and machine learning. My current research focuses on:

- Theoretical foundations of LLM alignment
- Mean-field high-dimensional statistics
- Combinatorial stochastic processes

Preprint.

- (1) [A Markov Chain Approach to Preference Alignment](#)
Takuya Koriyama and Tengyuan Liang
arXiv:2606.22652, 2026
- (2) [Asymptotic Inference for Exchangeable Gibbs Partition](#)
Takuya Koriyama
arXiv:2506.21527, 2025, minor revision at Stochastic Processes and their Applications
- (3) [Existence of solutions to the nonlinear equations characterizing the precise error of M-estimators](#)
Pierre C. Bellec and Takuya Koriyama (alphabetical)
arXiv:2312.13254, 2023, submitted.

Publication.

- (1) [Asymptotics of resampling without replacement in robust and logistic regression](#)
Pierre C. Bellec and Takuya Koriyama (alphabetical)
Bernoulli **32(4)**: 3063-3087, 2026.
- (2) [Denoising Diffusions with Optimal Transport: Localization, Curvature, and Multi-Scale Complexity](#)
Tengyuan Liang, Kulunu Dharmakeerthi, and Takuya Koriyama
Transactions on Machine Learning Research, 2835-8856, 2026.
- (3) [Precise asymptotics of bagging regularized M-estimators](#)
Takuya Koriyama, Pratik Patil, Jin-Hong Du, Kai Tan, and Pierre C. Bellec
The Annals of Statistics, **54(2)** 796-833, 2026.
- (4) [Asymptotic mixed normality of maximum likelihood estimator for Ewens–Pitman partition](#)
Takuya Koriyama, Takeru Matsuda and Fumiyasu Komaki
Advances in Applied Probability, **58(1)** 214-234, 2026.
- (5) [Phase transitions for the existence of unregularized M-estimators in single index models](#)
Takuya Koriyama and Pierre C. Bellec
Proceedings of the 42nd International Conference on Machine Learning (ICML 2025).
- (6) [Error estimation and adaptive tuning for unregularized robust M-estimator](#)
Pierre C. Bellec and Takuya Koriyama (alphabetical)
Journal of Machine Learning Research **26 (16)**, 1-40, 2025.
- (7) [Corrected generalized cross-validation for finite ensembles of penalized estimators](#)
Pierre C. Bellec, Jin-Hong Du, Takuya Koriyama*, Pratik Patil* and Kai Tan (alphabetical)
Journal of the Royal Statistical Society: Series B, **87(2)**, 289-318, 2025.

- (8) [Fully Data-driven Normalized and Exponentiated Kernel Density Estimator with Hyvärinen Score](#)
Shunsuke Imai, Takuya Koriyama, Shouto Yonekura, Shonosuke Sugawara and Yoshihiko Nishiyama
Journal of Business & Economic Statistics 43 (1), 110-121, 2025.

Award.

- IMS Hannan Graduate Student Travel Award, 2025.
- Travel award, Yale FDS workshop honoring Andrew Barron, 2024.
- Best Ph.D. Qualifying Exam Performance, Department of Statistics, Rutgers University, 2023.
- Student Travel Award, IMS International Conference on Statistics and Data Science, 2022.
- Best Presentation Award, 16th Japan Statistical Society Spring Meeting, 2022.
- Japan Statistical Society Certificate Director's Award, 2022.

Talk.

- Proceedings of the 43rd International Conference on Machine Learning (ICML), Seoul, July 2026.
- ISBA satellite Meeting 2026, Tokyo, July 2026.
- Proceedings of the 42nd International Conference on Machine Learning (ICML), Vancouver, July 2025.
- Workshop on Statistical Network Analysis and Beyond (SNAB), Tokyo, June 2025.
- IMS International Conference on Statistics and Data Science, Nice, December 2024.
- Workshop Honoring Andrew Barron: Forty Years at the Interplay of Information Theory, Probability and Statistical Learning, Yale University, April 2024.
- IMS Asia Pacific Rim Meeting, Melbourne, January 2024.
- Seminar talk, Graduate school of Economics, the University of Tokyo, February 2023.
- IMS International Conference on Statistics and Data Science, Florence, December 2022.
- Workshop on random partition of integers, the University of Tokyo, June 2022.
- Japan Statistical Society Meeting, Keio University, March 2022.

Review.

- Information and Inference a Journal of the IMA, Journal of the American Statistical Association, Annals of Statistics